Sceen Shot

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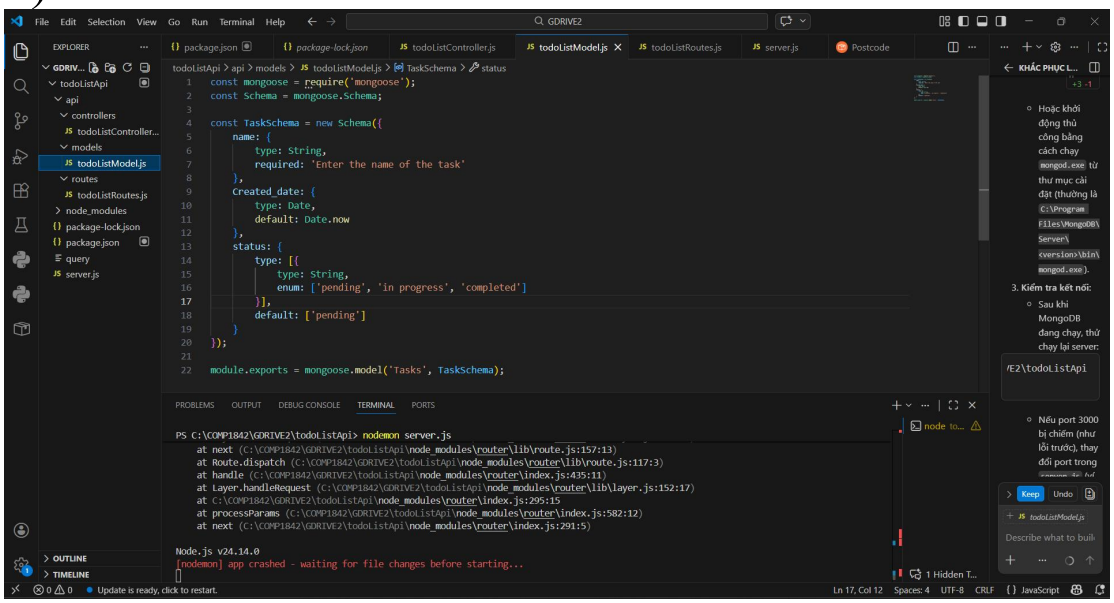

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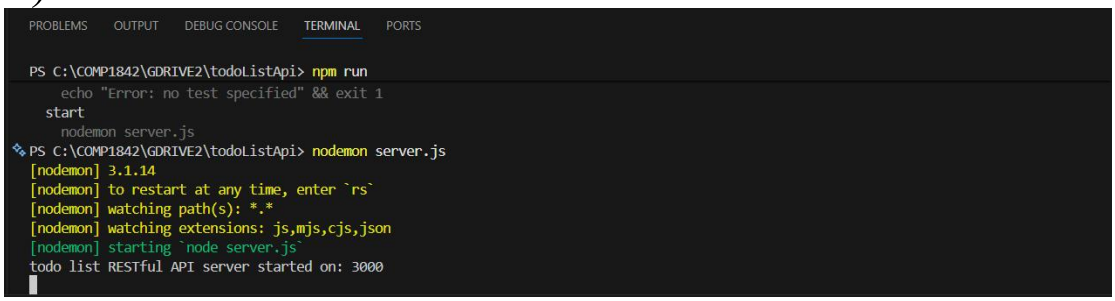

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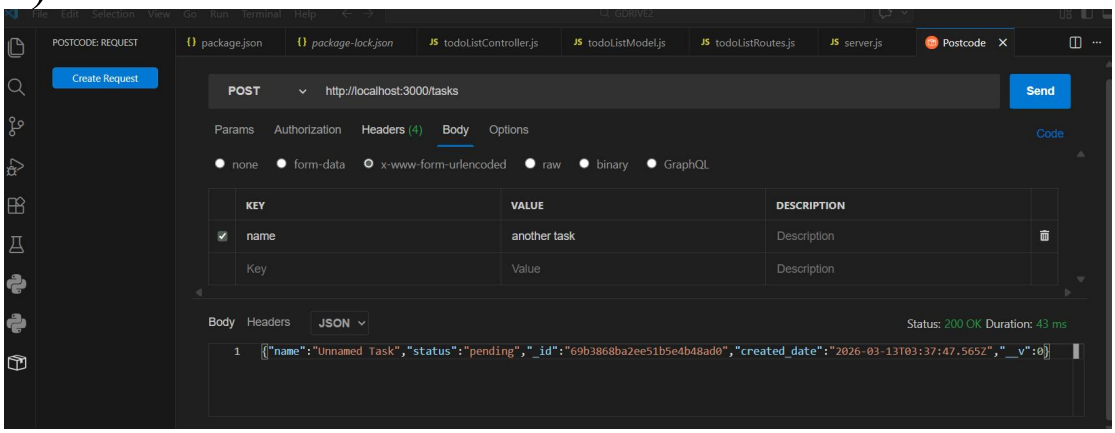

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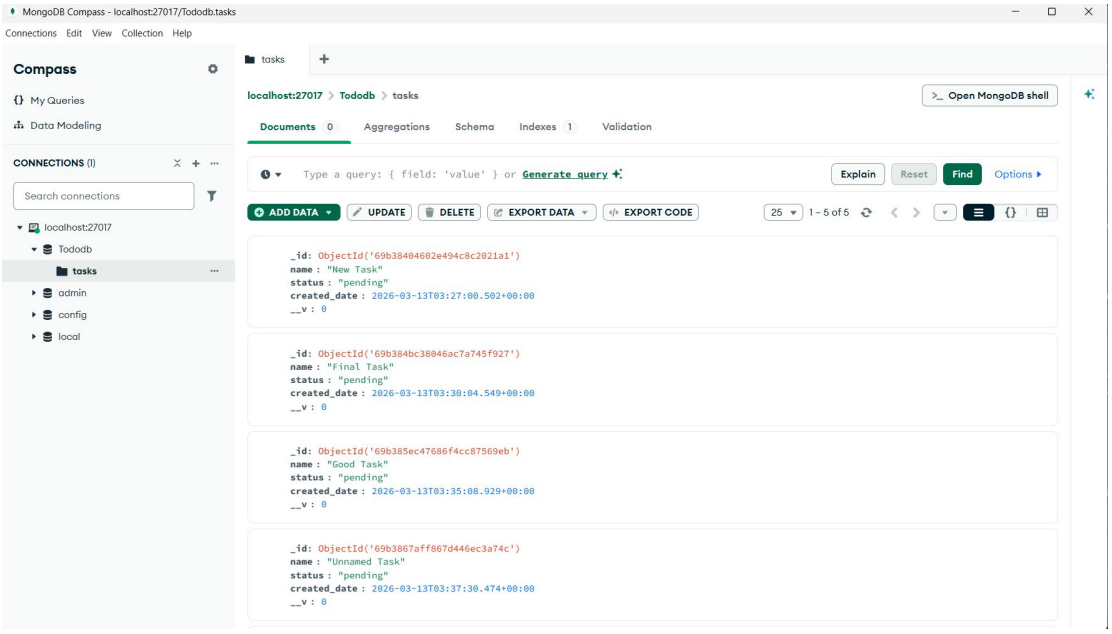

Understanding

Understanding RESTful APIs and CRUD operations is essential when building modern web applications. An API (Application Programming Interface) acts as a communication bridge between two software systems, allowing them to exchange data through defined requests and responses. In web development, APIs often follow the REST architectural style, which uses HTTP methods to perform operations on resources. A resource can be any data object exposed by the server, such as a task, user, or word in a database.

In a RESTful API, communication happens between a client and a server. The client sends a request to a specific endpoint, usually in the form of a URL, and specifies the action using an HTTP method such as GET, POST, PUT, or DELETE. The server processes the request and returns a response, typically formatted as JSON data. One important concept of REST is statelessness, meaning the server does not store client information between requests. Each request must contain all necessary information for the server to process it.

CRUD operations are closely related to REST APIs. CRUD stands for Create, Read, Update, and Delete, which correspond to the HTTP methods POST, GET, PUT, and DELETE. These operations allow developers to manage data in a database through an API. In a Node.js application using Express and MongoDB, models define the structure of the data, routes define the endpoints, and controllers implement the functions that interact with the database using libraries such as Mongoose.

Overall, RESTful APIs provide a structured and scalable way for frontend applications, such as Vue, to communicate with backend services and manage data efficiently.